

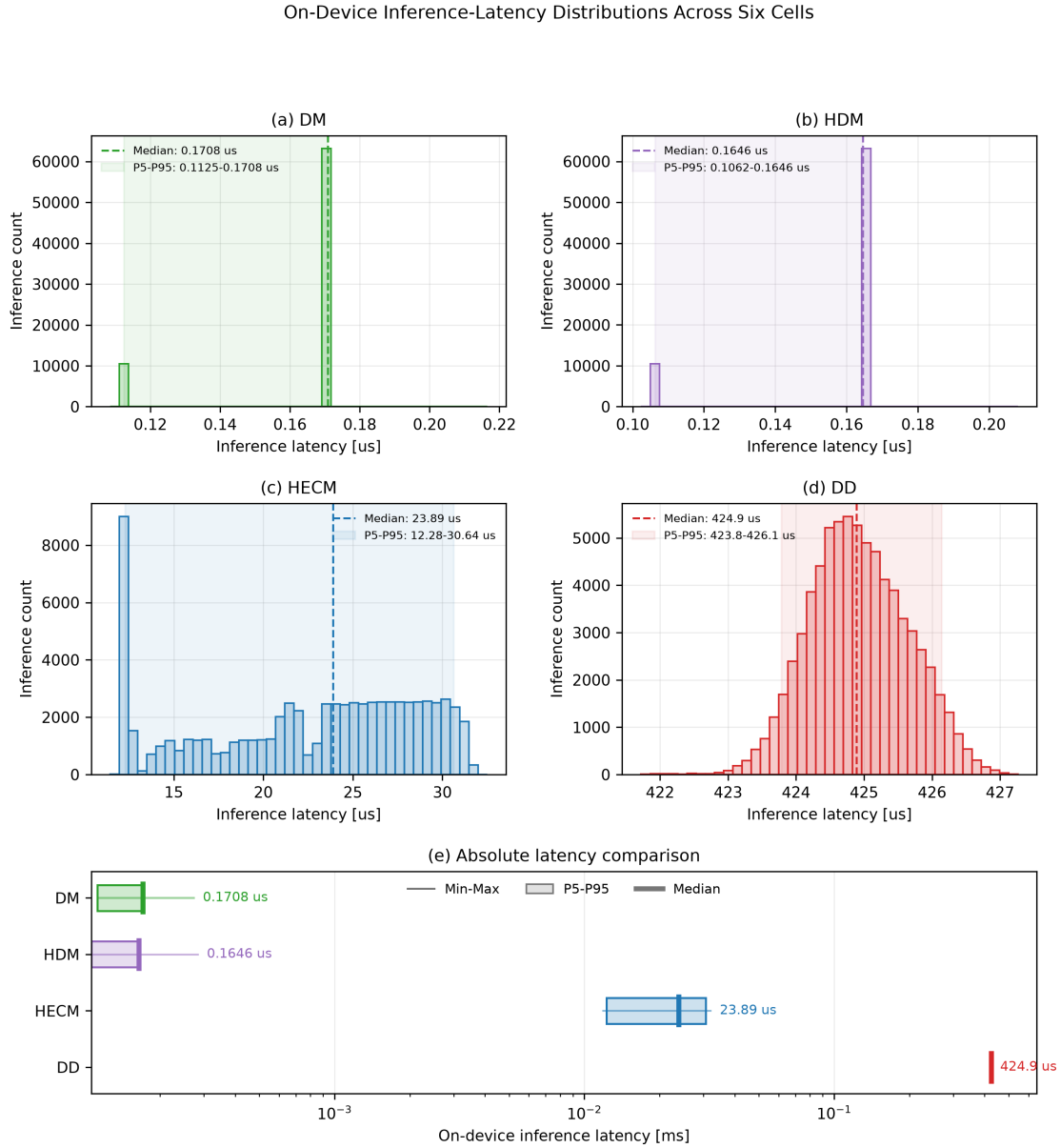


Figure 6.18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a) to (d) show per-inference distributions. Panel (e) compares the minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model comparison.

The compiled DM, HDM, HECM, and continuous-state DD firmware occupies 27.0, 27.1, 470.1, and 115.9 KiB of flash. Measured peak runtime RAM is 2.4, 2.4, 2.5, and 4.1 KiB, respectively. Figure 6.19 compares both quantities. HECM flash is dominated by its parameter surfaces, whereas neural parameters account for most of the DD flash demand. All four isolated SOC implementations fit within